

Design of Binary Architecture for Successive Approximation Analog-to-Digital Converter

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Abstract—The design of comparator, 4-bit binary capacitive-array Digital-to-Analog Converter (DAC) and Successive Approximation Register (SAR) for Successive Approximation Analog-to-Digital Converter (SA-ADC) is implemented in this paper. The comparator contains a pre-amplifier followed by a regenerative latch and an SR latch. The capacitive-array DAC itself does the functioning of sample and/or hold (S/H) circuit i.e. an explicit sample and/or hold (S/H) circuit is not required. The binary-weighted capacitor DAC uses a switching technique to charge the capacitors by the charge distribution principle. For low power purpose, the flip flops used in the SAR are set-reset D-flip flops. The main advantage of this SA-ADC architecture is its low power consumption due to inherent S/H operation inside the capacitor DAC. The supply voltage of the circuits is 700mV. The simulation results of a 4bit binary capacitor DAC, SAR and the comparator circuit is shown on 90nm CMOS technology using Tanner EDA Tools.

Index Terms—Binary capacitor-based Digital-to-Analog Converter (DAC), Capacitive-array DAC, Comparator, Sample/Hold (S/H) circuit, Successive Approximation Analog-to-Digital Converters (SA-ADCs), Successive approximation Register (SAR)

I. INTRODUCTION

Successive approximation analog-to-digital converter (SA-ADC) has become very attractive in low power applications due to their low power consumption. As shown in Fig. 1 [1], [2], a conventional SA-ADC consists of a sample-and-hold (S/H) circuit, comparator, Successive Approximation Register (SAR), and Digital-to-Analog Converter (DAC) [1], [2]. The operation of SA-ADC is also follows. The input voltage (V_{IN}) is sampled by the S/H circuit during the sampling phase & produces an output V_H , this voltage does not change during the conversion phase [1], [2]. During the

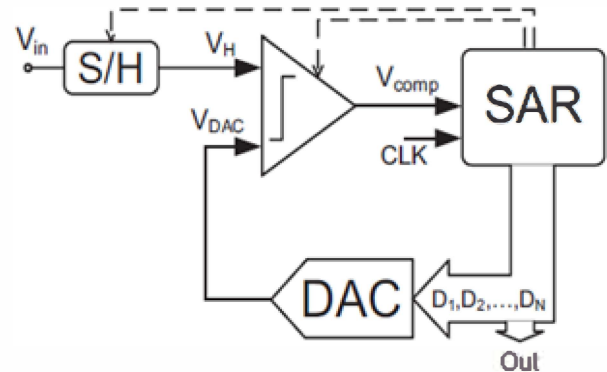


Fig. 1. Successive approximation ADC block diagram

conversion phase, the output of DAC (V_{DAC}) is compared to V_H and the result of comparison is fed to Successive Approximation Register (SAR). A SAR is designed to perform the binary search algorithm to determine the value of the bits sequentially based on the result of the comparator and supply an approximate digital code of V_{IN} to the DAC [1], [2]. After N clock cycles in the conversion phase, all N bits of the digital word are obtained [1], [2].

In an SA-ADC structure, one of the most important building blocks is the Digital-to-Analog Converter (DAC). The DAC determines the accuracy and the conversion speed of the converter. It also consumes major part of overall power of the DAC [1], [2]. The DAC used in SA-ADC can be realized in various ways; e.g. capacitor-based DAC, switched-current DAC, or R-2R ladder DAC [1], [2]. Among these architectures, the capacitor-based DAC has become more popular because of zero quiescent current. Also, in most technologies, resistor mismatch and variation are greater than those of a capacitor [1], [2].

The capacitive-array DAC used in SA-ADC can be implemented in three different ways, i.e., binary-weighted, unary-weighted, and segmented capacitive array [1], [2]. Binary-weighted arrays are simple for implementation and do not suffer from design complexity and also has no power consumption overhead due to the thermometer decoder which is required in unary and segmented capacitive array [1]. The binary-weighted capacitive-array DAC have different structures, e.g., conventional capacitive array, split capacitive array, capacitive array with attenuation capacitor, C-2C capacitive array, SA-ADC with dual capacitive array,

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capacitive array with merged capacitor switching, capacitive array with monotonic switching method, and charge redistribution DAC with step-wise charging [1], [2].

The rest of this paper is organized as follows. In Section II, The design of comparator, binary weighted capacitor array DAC and SAR are presented along with their working. Section III includes the simulation results of the designs. Finally, the conclusion is given in Section IV.

II. PROPOSED WORK

The conventional structure of SA-ADC structure employing binary capacitor array DAC is shown in Fig. 2. The S/H function can be realized either by using an explicit S/H circuit [1], [3] or by the capacitive array DAC itself. In this paper, explicit S/H circuit is not used and the sampling operation is performed by capacitor DAC itself.

A. Comparator

The comparator used in SA-ADC plays an important role. The main purpose of the comparator is to compare a signal with a reference signal and produce an output depending on whether the input signal is greater or smaller than the reference signal. Fig. 3 depicts the design of the comparator circuit [1]. The comparator circuit consists of three blocks a pre-amplifier, a regenerative latch and a SR latch. A pre-amplifier is followed by a regenerative latch and a SR latch [1], [4]. A sufficient gain is provided by the input pre-amplifier to compensate for the relatively high input referred offset voltage of the latch and for isolation from latch kickback noise [1], [4]. In order for the pre-amplifier to operate correctly, all the devices are biased in the triode region and all the pre-amplifier transistors biased in weak inversion region [1], [4].

The comparator has two inputs V_{in+} & V_{in-} , where V_{in+} is sinusoidal input signal compared with referenced voltage V_{in-} and depending on whether V_{in+} is greater than or smaller than V_{in-} output voltage is determined.

- $$\begin{aligned} \text{If } V_{in+} - V_{in-} > 0 \text{ then } V_{out} &= "1" \\ \text{If } V_{in+} - V_{in-} < 0 \text{ then } V_{out} &= "0" \end{aligned} \quad \begin{matrix} (1) \\ (2) \end{matrix}$$

B. Binary weighted capacitor array DAC

The Digital to Analog Converter (DAC) converts the digital value at the output of the SAR logic to the analog value. Then this value is compared to the input signal with the help of comparator. Fig. 4 shows an N bit binary capacitive DAC. The value of each capacitor in the DAC is given by

$$C_i = 2^{N-i} C_u \quad (3)$$

where C_u is the capacitance of unit capacitor, N is the total number of bits and i is the index number of the capacitor [1]. The last capacitor C_u is a dummy capacitor and it has a value equal to the LSB capacitor [1], [5]. The capacitor array DAC uses a switching technique to charge the capacitors. There are three phases of operation to perform a conversion. First phase is the sampling phase. In this phase, the top plate switch is the

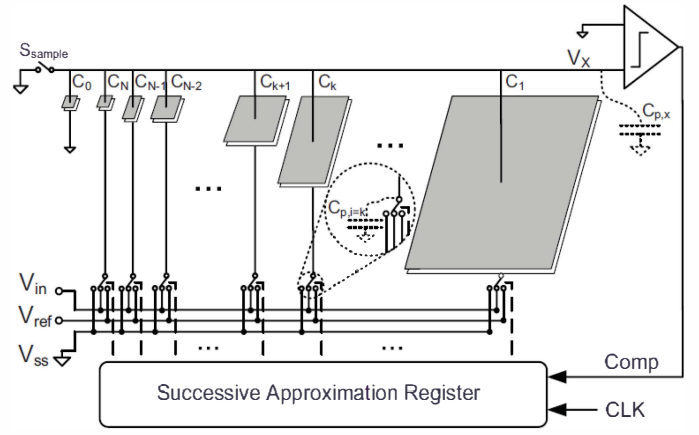


Fig. 2. SA-ADC structure employing binary capacitor DAC

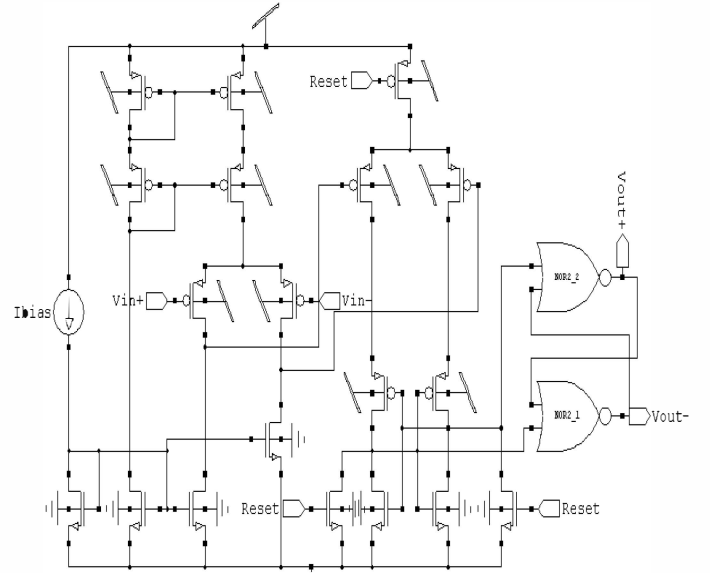


Fig. 3. Design of comparator

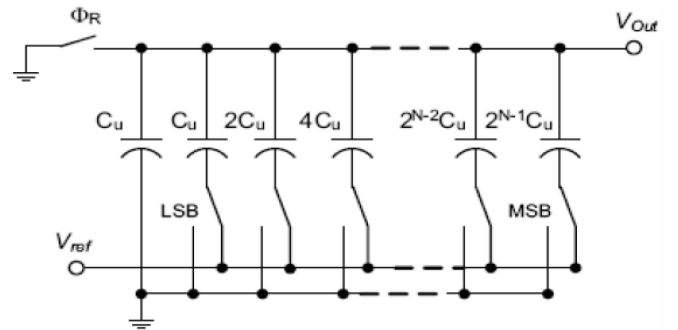


Fig. 4. N bit binary capacitive DAC

sampling phase. In this phase, the top plate switch connects the top plates of all the capacitors to the V_{CM} and the bottom plates are connected to V_{IN} . Thus, V_{IN} is sampled on the capacitor array [5].

During the hold phase, the top plate switch is opened and the bottom plate switch connects all the capacitor to ground and hence a charge of $-V_{IN} + V_{CM}$ is stored in the capacitor array [5].

During the redistribution phase, actual conversion is performed; the digital data determines the status of the bottom plate switch. According to the digital the position of the bottom plate switching is decided, whether the switch should be connected to the V_{REF} or ground. The output of the DAC is obtained in this stage [5]. Fig. 5 shows the design of 4-bit binary weighted capacitor DAC [5].

Fig. 6 shows the design of top plate switching [5]. During the sampling phase, both the PMOS switches are turned on and the value of the V_{CM} is obtained at the output. This charges the top plate of the capacitor. Table I shows the bottom plate switch operation [5]. Fig. 7 shows the design of bottom plate switching [5]. In this circuit, the bottom plate of the capacitors are charged to either V_{IN} , V_{REF} or ground depending on the sample signal. In sampling phase, the sample signal is high, and the analog input signal is sampled into the capacitor array. During the conversion phase, the sample signal is low, and depending on the digital data, the bottom plate of the capacitor is connected to either ground or V_{REF} i.e. if the data is logic 0, then the capacitor is connected to ground and if the data is logic 1, capacitor is connected to V_{REF} .

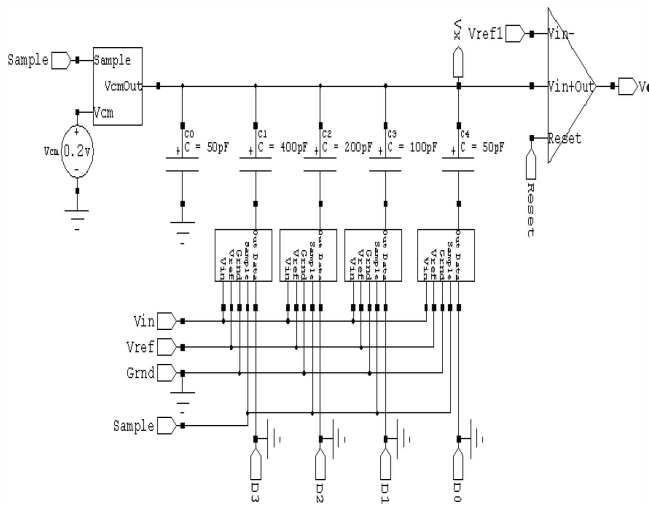


Fig. 5. Design of 4 bit binary capacitor DAC

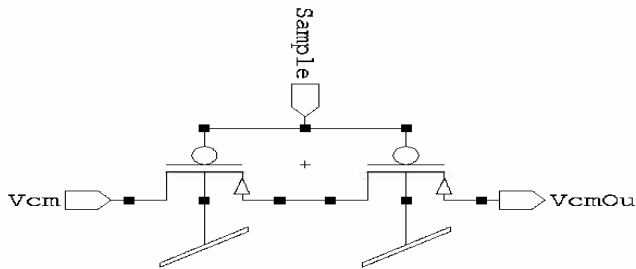


Fig. 6. Design of top plate switching

TABLE I
BOTTOM PLATE SWITCH OPERATION

Sample	Digital Data	V_{bottom_plate}
1	X	V_{IN}
0	1	V_{REF}
0	0	Ground

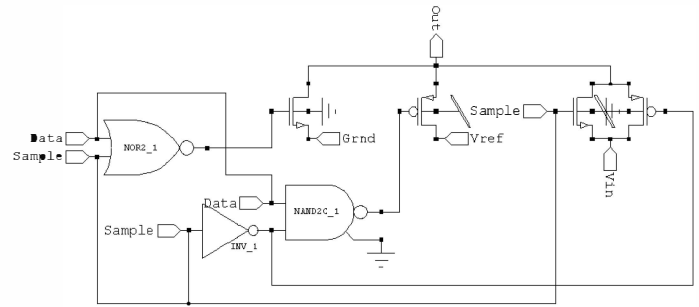


Fig. 7. Design of bottom plate switching

C. Successive Approximation Register

The Successive approximation register (SAR) circuit determines the value of each digital bit in a sequential manner, based on the output of the comparator [1]. The design for the SAR is shown in Fig. 8 [5], [6]. The SAR architecture consists of at least $2N$ flip flops. It consists of chain of N flip flops of sequencer and another chain of N flip flops as a code register. A sequencer is a shift register and is used to guess the digital bits and code register are used to store the results [1].

For a 4 bit SAR design, each conversion takes 6 clock cycles. In the first clock cycle, SAR is in the reset mode and all the outputs are zero. In the next 4 clock cycles, data is converted and each bit is determined sequentially. In this clock cycles, the most significant flip flop is set to one which corresponds to MSB of the DAC. Then the counter shifts '1' through the flip flops from MSB to LSB [5].

In each clock cycle, one of the outputs in the sequencer sets a flip flop in the code register. The output of this flip flop which is set by the sequencer is used as a clock signal for the previous flip flop [5]. The last cycle is for storing the results of the complete conversion [5].

The set-reset D-flip flops are used in this structure. For low power purpose, transmission gate based flip flops are used [5]. The design of the D-flip flop is shown in Fig. 9. The advantages of the SAR architecture are that, since the logic is iterative, it can be extended to higher resolutions by just extending the shift registers [5]. Furthermore, the total signal transitions are kept low, and, hence, the energy consumption is minimized [5], [6]. In order to decrease the power consumption even more, the supply voltage can be reduced [5], [6].

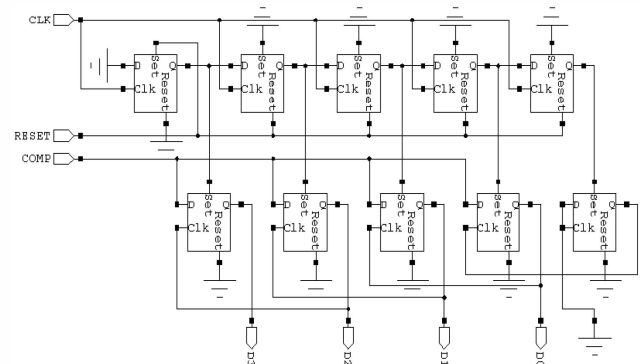


Fig. 8. Design of successive approximation register

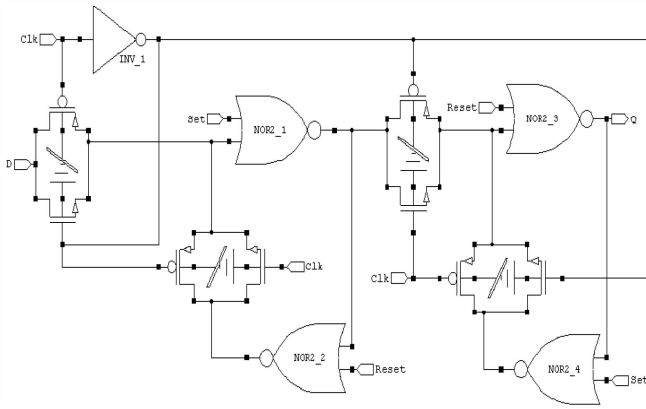


Fig. 9. Design of D-flip flop

III. SIMULATION RESULTS

All the structures are designed and simulated using 90nm CMOS technology using Tanner EDA tool, with supply voltage of 700mV, 4 bit capacitor DAC with $C_u = 50\text{pf}$.

Fig. 10 depicts the transient response of the comparator. A sinusoidal input of 700mV is applied and is compared with a reference signal of 500mV. If the input voltage is greater than 500mV, the output shows a positive pulse, else the output is zero. The average power consumption of this circuit is 0.25 μW and power consumption due to supply voltage is 19.586 μW .

Fig. 11 shows the transient response of 4 bit binary capacitor array DAC for input 0000. The capacitance of a unit capacitor is 50pf. The capacitances of other capacitors are calculated depending on this value. The input voltage supplied to this circuit is 200mV. The average power consumption of this circuit is 25.14 μW and the power consumption due to supply voltage is 49.19 μW .

Fig. 12 shows the transient response for a 4-bit SAR circuit. All the data bits are shifted one by one in every clock pulse. After every 6 clock pulse, the conversion is repeated. The average power consumption of this circuit is 23.74 μW and the power consumption due to supply voltage is 620.20 nW.

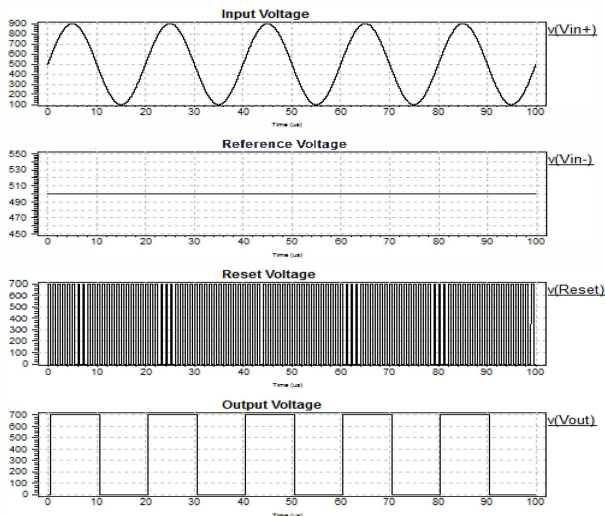


Fig. 10. Transient response of comparator

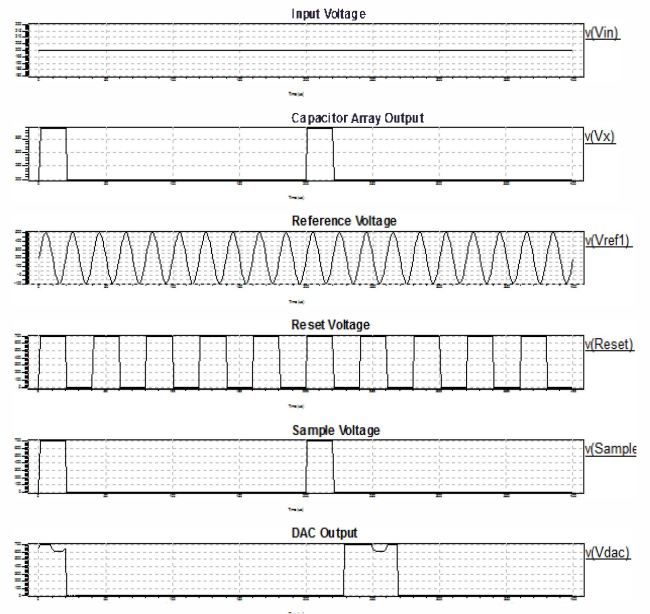


Fig. 11. Transient response of binary DAC

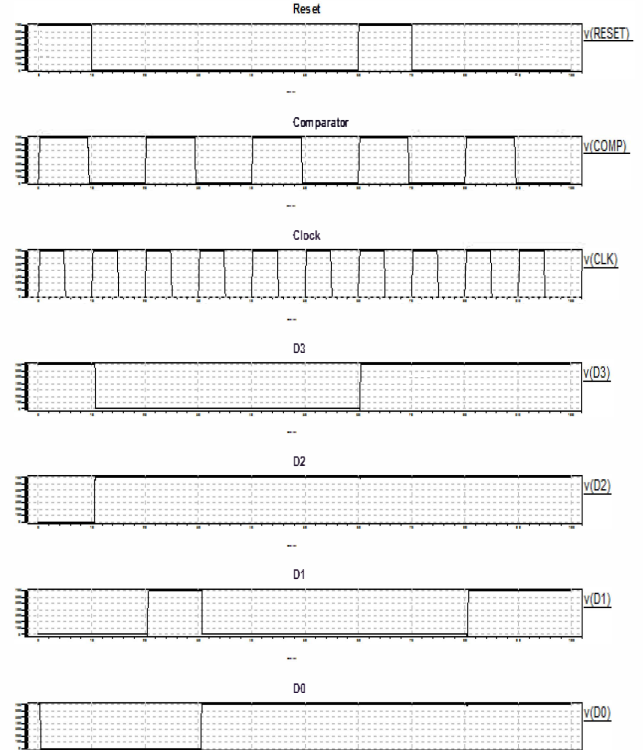


Fig. 12. Transient response of successive approximation registers

IV. CONCLUSION

A comparator, binary capacitive DAC and SAR circuit is designed for the SA-ADC architecture. A comparator does not depend on the number of bits of the capacitor array and this circuit can be employed with similar power consumption for any architecture of capacitor-array DAC. Furthermore, bandwidth of the comparator is improved. Binary DAC is easy for implementation, it has no design complexity and it has no

power consumption overhead due to binary to thermometer decoder, which is required in unary and segmented capacitor DAC. Using a switching technique, the use of sample and hold circuit can be avoided, and the capacitor DAC functions as S/H circuit. By using this SAR architecture power consumption is minimized.

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